MAGNETOM

Function Description System

Power Distribution System

MAGNETOM ESSENZA

Document Version

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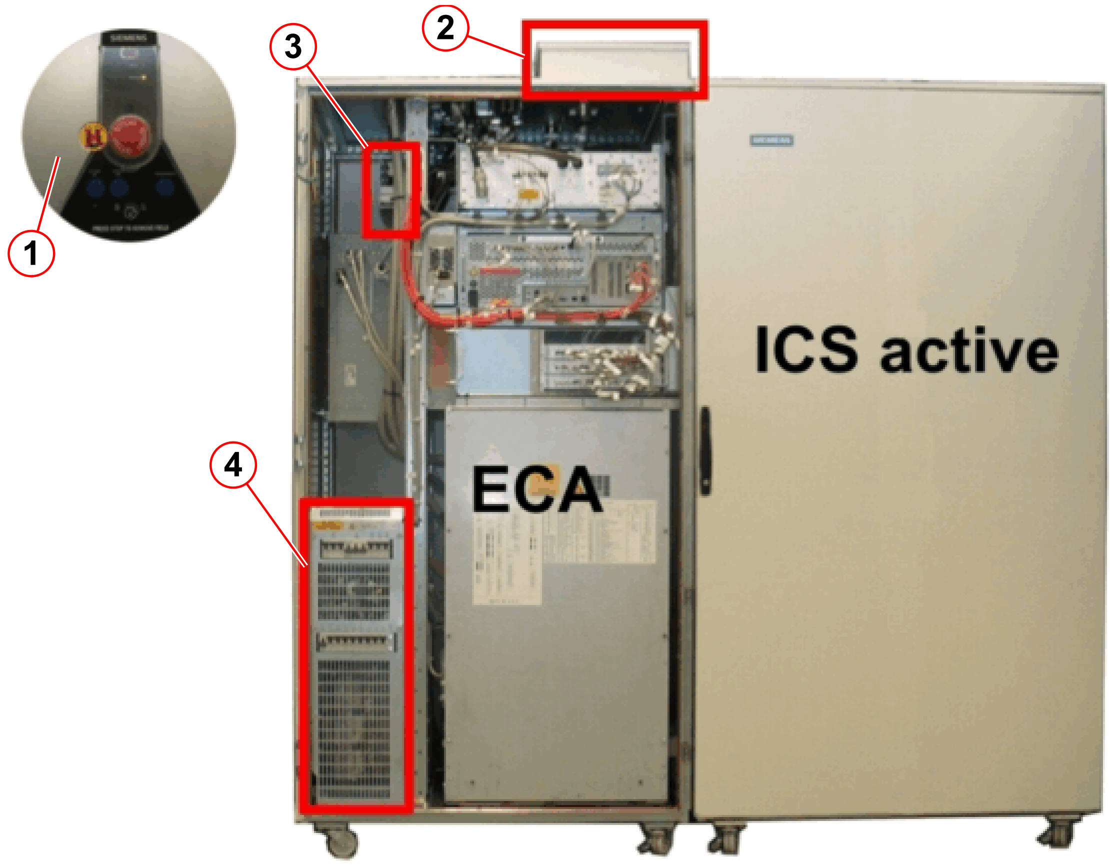
1.1 Task of the Power Distribution System (PDS)

The Power Distribution System (PDS), which sometimes is called the line power distribution (LPD), provides the system components with line power and the necessary circuit breaker protection. Some of its additional functions are net filtering, overtemperature protection, and alarm box connections.

The following components are part of the PDS:

1. Alarm Box
2. Mains Connection Box with system circuit breakers
3. 24/36 V power supply
4. PDS section in the ECA with component circuit breakers, main transformer T1, contactors, and relays.

Fig. 1: PDS components



1.1.1 Alarm Box

The Alarm Box is usually mounted next to the MRC in the control room. Its display functionality for the magnet monitoring is described in the magnet part of the FUN. Only the power-up and power-down functions are shown in the following description.

Fig. 2: Alarm Box

(1) POWER-LED indicates supply to power-up circuitry
(2) Key switch
(3) SYSTEM ON button
(4) SYSTEM ON LED
(5) SYSTEM OFF button

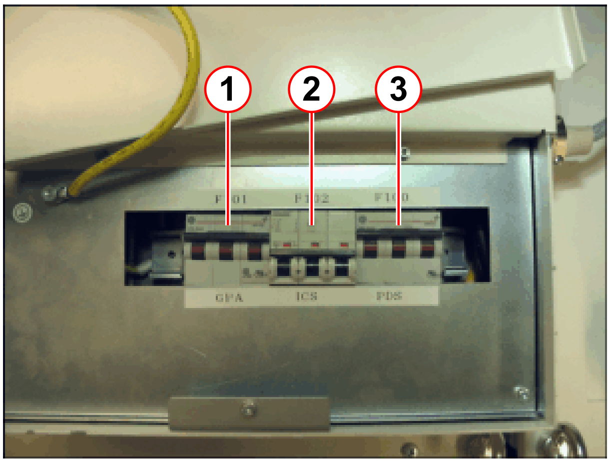
The Alarm Box has limited functions compared to the one from MAGNETOM Avanto/ Espree systems. It has no CAN-interface, no stand-by button, and just one red FAULT-LED indicating problems with the magnet and the cooling system.

> **i** If the key switch is in the lock position and removed, you can still switch on the system using the ON-button inside the ECA.

1.1.2 Mains Connection Box (MCB)

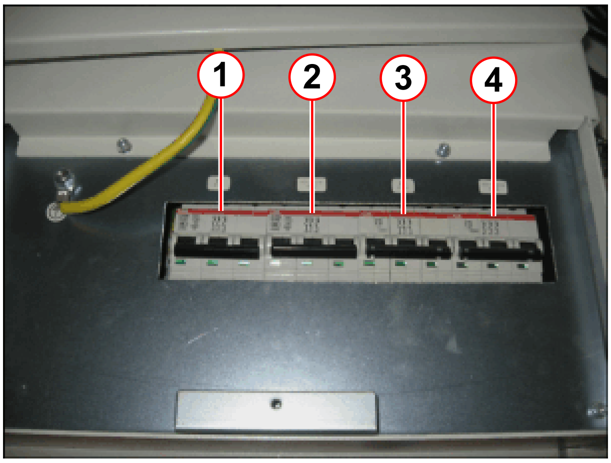
The MCB is located on the top right corner of the ECA. It is split up into two parts. In the back part you find the terminal point X100 for the mains supply from the customer. In the front part of the MCB you can find the three/four system circuit breakers.

Fig. 3: System circuit breakers in MCB (old type)



(1) F101 (GPA power stage supply)
(2) F102 (ICS active or passive)
(3) F100 (PDS in ECA)

Fig. 4: System circuit breakers in MCB (new type)



(1) F101 (GPA power stage supply)
(2) F102 (ICS active)
(3) F100 (PDS in ECA)
(4) F103 (ICS passive)

> The MCB of the first systems (system serial number < 38028) contain only three circuit breakers; later versions have four. There is a separate circuit breaker for each ICS.

1.1.3 24/36 V Power Supply

Fig. 5: 24/36 V Power Supply

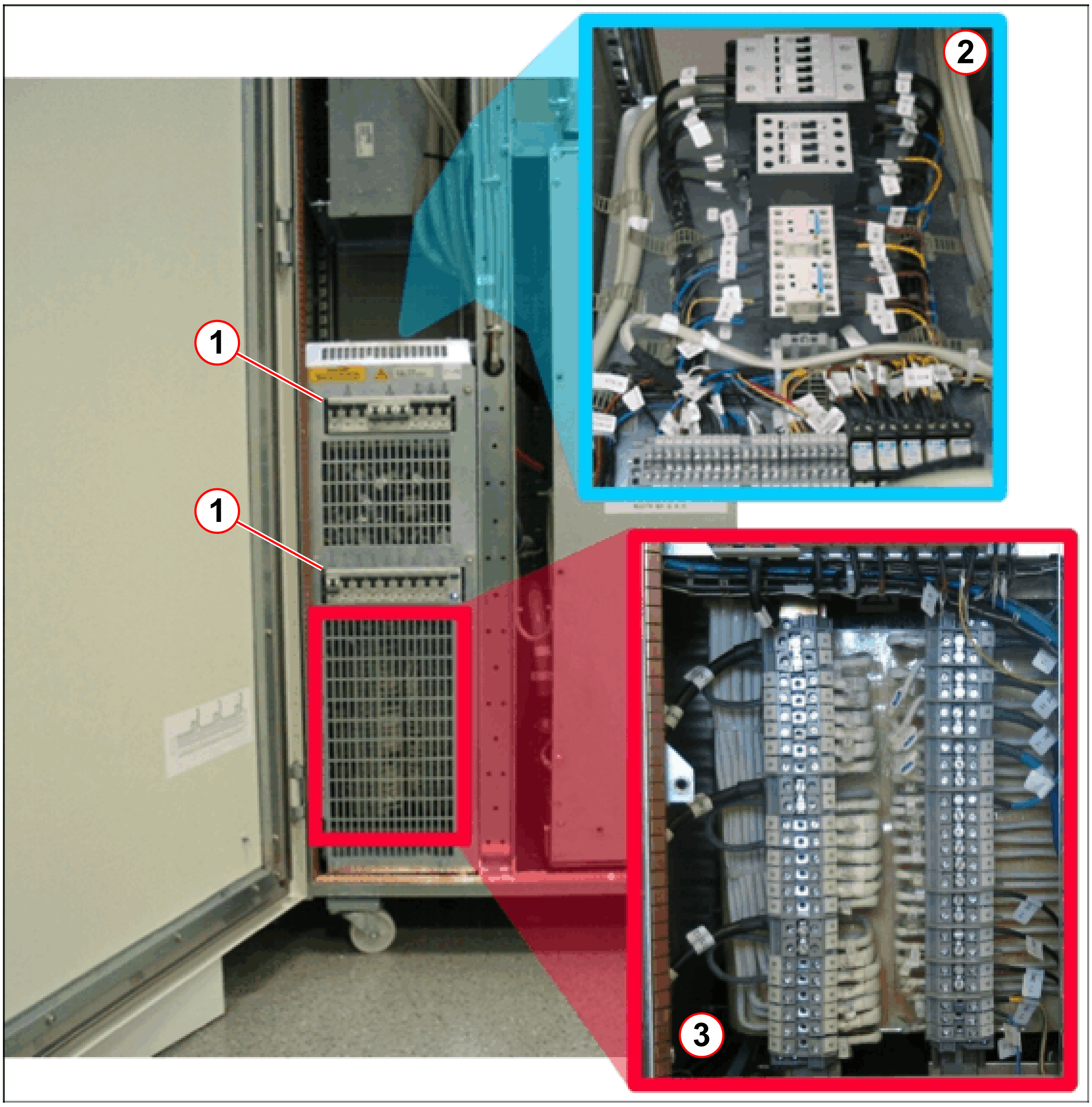
The 24/36 V power supply supplies the following components:

- 24 V for the power-up circuitry
- 36 V for the MSUP and the Alarm Box
- 24 V for the CANopen bridge (COB)

1.1.4 PDS section inside the ECA

The component circuit breakers are located in the PDS section of the ECA. It also contains all contactors, relays, and the main transformer T1.

Fig. 6: PDS section



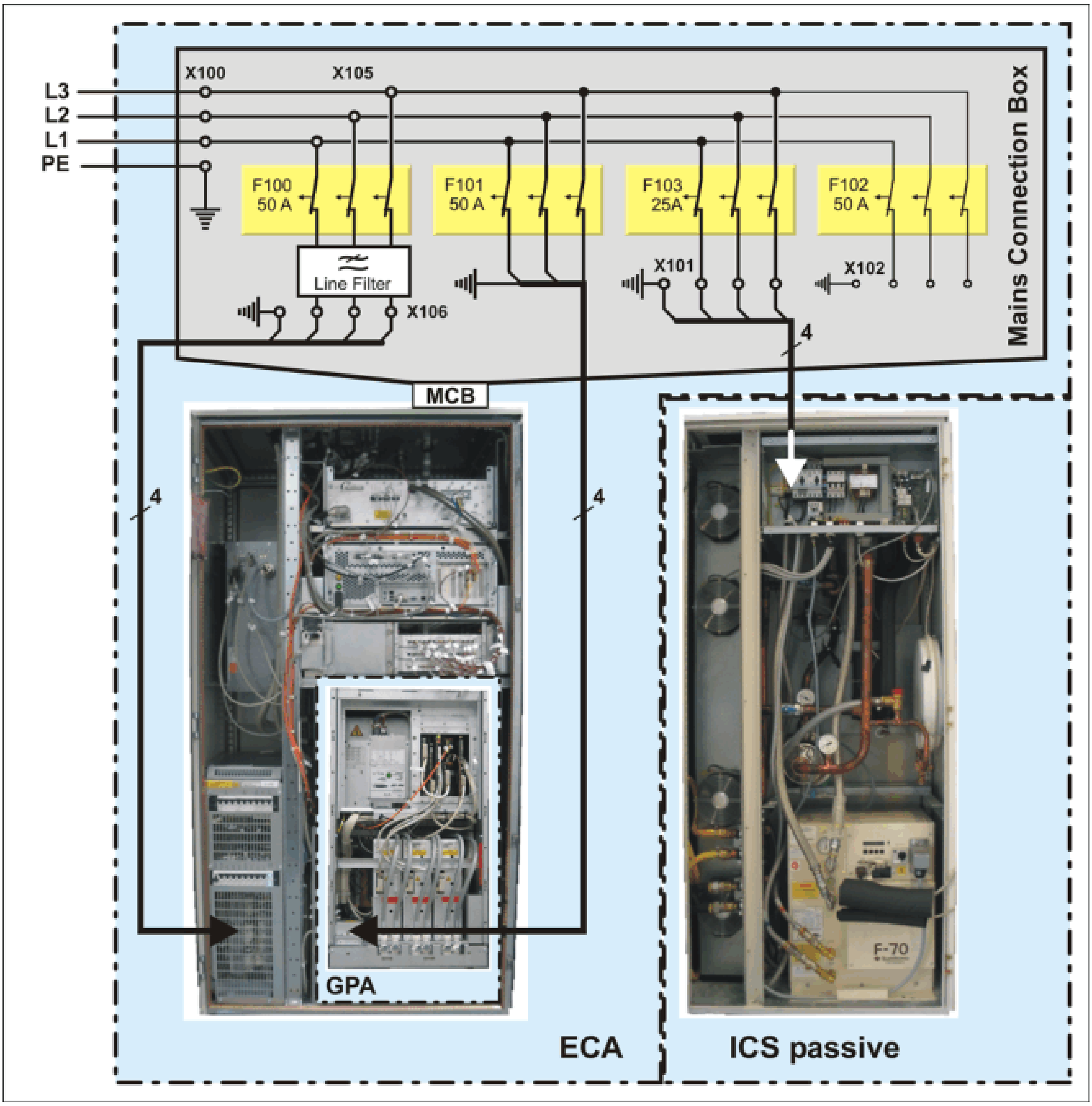
(1) Component circuit breakers
(2) Contactors and relays
(3) Main transformer T1

1.2 PDS with passive ICS

The mains supply (L1, L2, L3, PE) from our customer is connected to X100 in the Mains Connection Box (MCB). From there the power supply is distributed to the MR system components in the following way:

- Circuit breaker F100 supplies most of the system components via main transformer T1.
- Circuit breaker F101 supplies the transformer T1 inside the GPA for the final stages.
- Circuit breaker F103 supplies the ICS passive.

Fig. 7: System circuit breakers with ICS passive

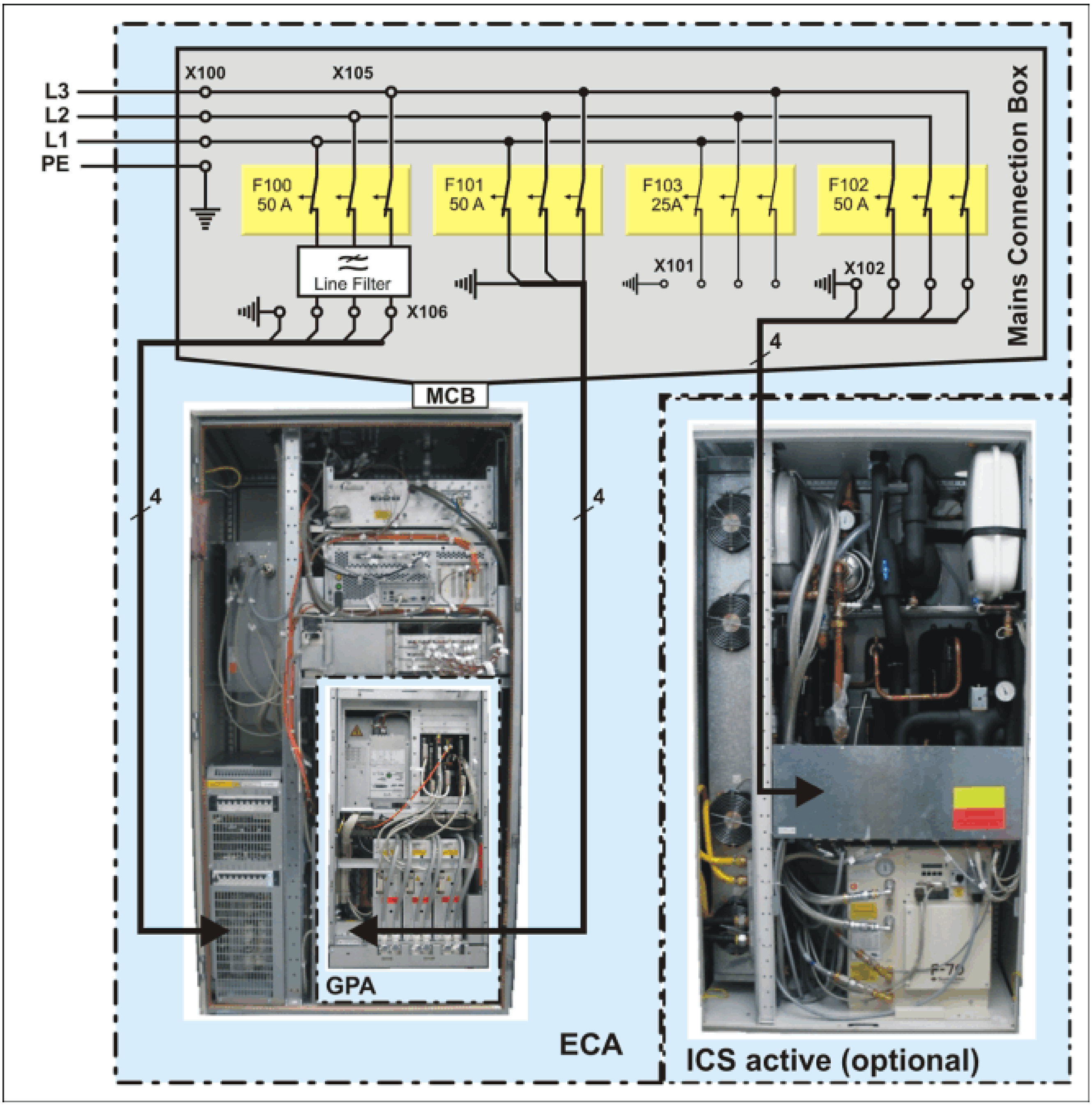


1.3 PDS with active ICS

The mains supply (L1, L2, L3, PE) from our customer is connected to X100 in the Mains Connection Box (MCB). From there the power supply is distributed to the MR system components in the following way:

- Circuit breaker F100 supplies most of the system components via main transformer T1.
- Circuit breaker F101 supplies the transformer T1 inside the GPA for the final stages.
- Circuit breaker F102 supplies the ICS active (optional cooling system).

Fig. 8: System circuit breakers with ICS active



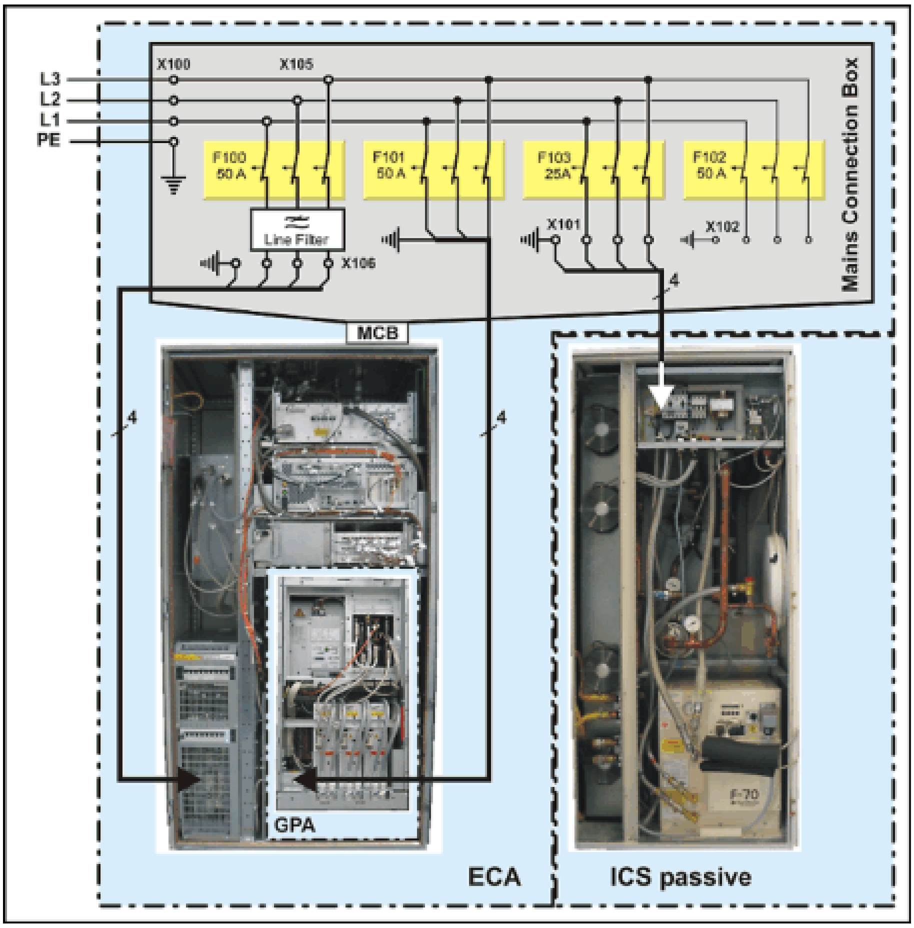
2.1 Introduction

2.1.1 Operating modes

The PDS switch logic can switch the MR scanner into three different operating modes:

- **System On** - All MR system components are switched on. The system may be used for imaging.
- **System Off** - Most of the MR system components are switched off.
- **System Standby** - All measurement-related components of the MR system are switched off to save energy. In that mode, only the MRC remains on, so that patient image evaluations are still possible.

Fig. 9: System circuit breakers



⚠ CAUTION

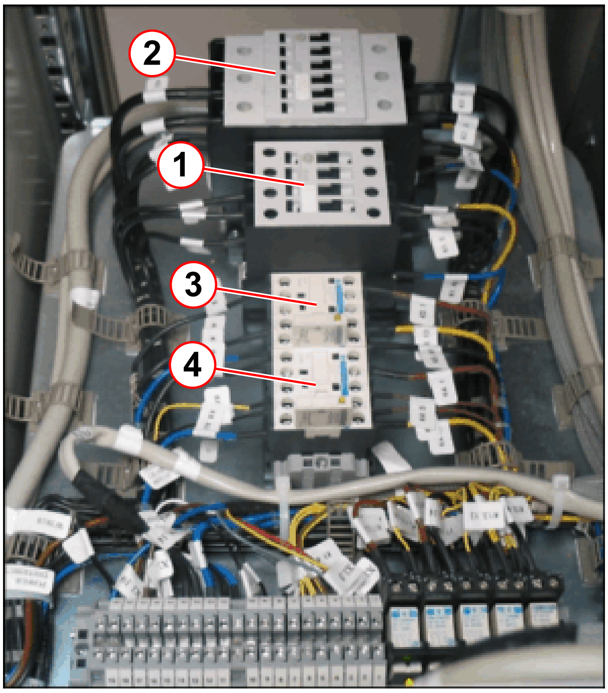
There is still power in the ICS, GPA power stage supply, MSUP, and power-up circuitry even when the system is switched off!

» Before beginning any troubleshooting procedures, always switch off the corresponding system circuit breaker in the Mains Connection Box (MCB).

2.1.2 Power switching

The PDS circuitry consists of four main contactors that switch the following system components via the corresponding component circuit breaker (CB):

Fig. 10: System contactors



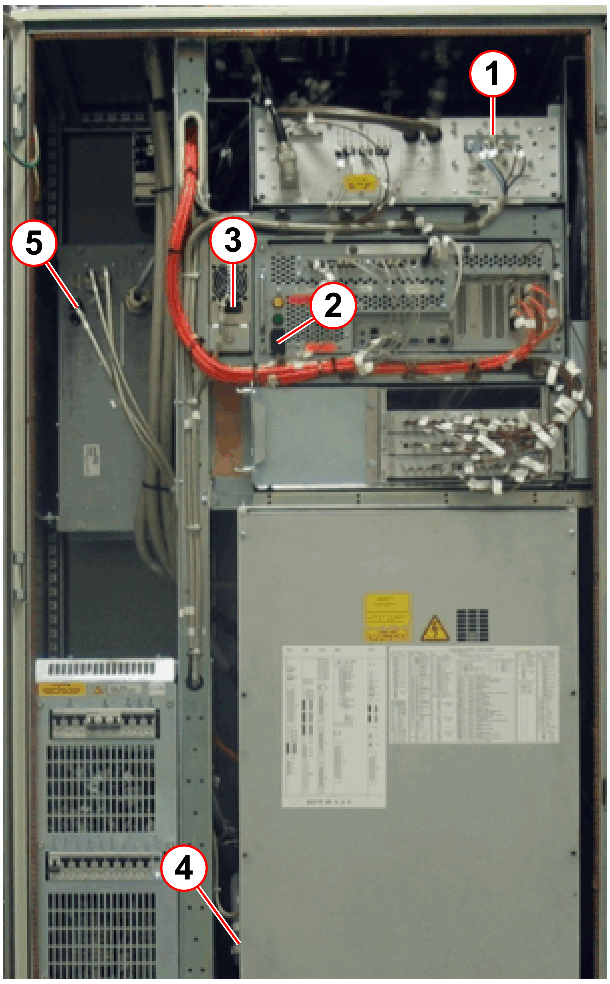
(1) K1
(2) K2
(3) K3
(4) K4

Tab. 1 contactors

contactor	CB	connector	component
K1	F11	X300	RFPA

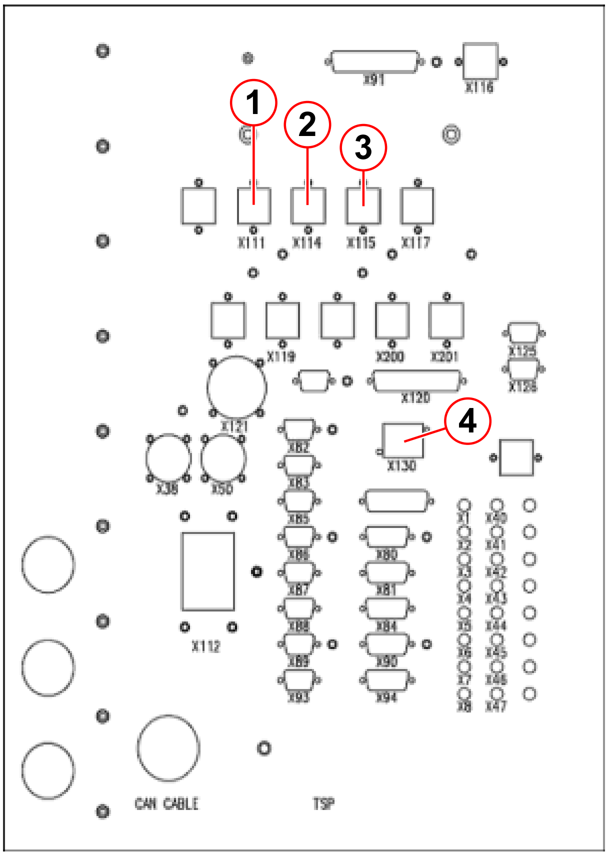
contactor	CB	connector	component
K2	F21	X700	E-Shim
	F23	X303	RFSU
	F26	X304	GSSU
	F32	X116	spare (not used)
	F22	X115	PTAB
	F24	X111	RFIS_60
	F28	X130	MPS (only during ramping)
K3	F31	X114	MRC
K4	F29	X302	MARS computer

Fig. 11: Power connectors in ECA



(1) X300 (RFPA)
(2) X302 (Mars)
(3) X303 (RFSU)
(4) X304 (GSSU)
(5) X700 (E-Shim)

Fig. 12: Power connectors on ECA roof panel



(1) X111 (RFIS_60)
(2) X114 (MRC)
(3) X115 (PTAB)
(4) X130 (MPS)

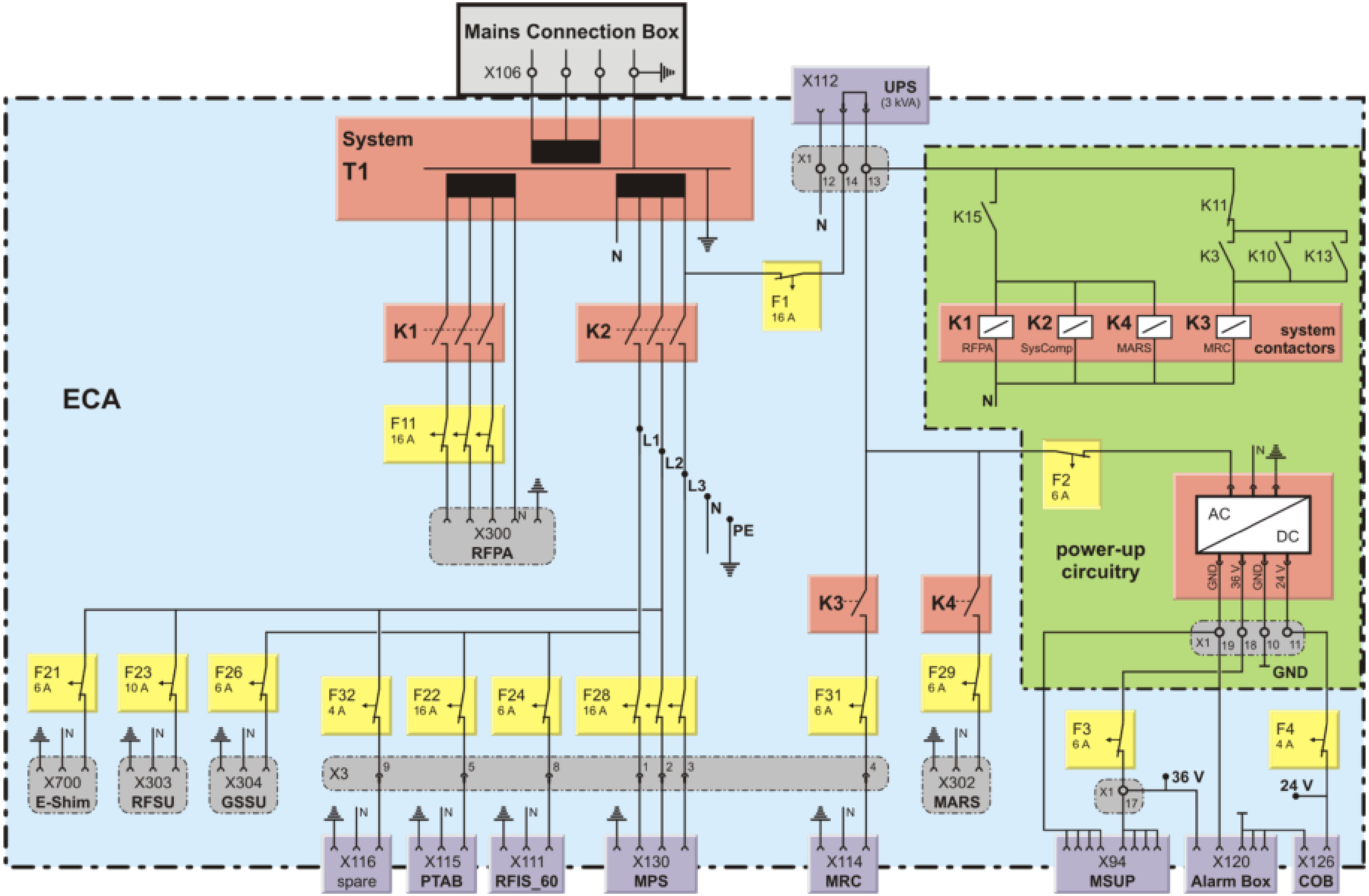
The system contactors K1 (RFPA), K2 (system components), and K4 (MARS) are always switched together via relay contact K15. The system contactor K3 for the MRC is switched in a separate way using relay contact K10. This makes it possible to keep the host running (only in standby mode).

The power-up circuitry is continuously supplied by circuit breaker F1. A shorting plug on X112 on the ECA roof panel could be used to connect a low-power UPS (3 kVA) for the two computer systems, Host and MARS.

Circuit breaker F2 supplies a DC power supply which delivers 24 Vdc and 36 Vdc for the dc-relays in the power-up circuitry and the supply for the CANopen bridge (COB).

The magnet supervision (MSUP) and the Alarm Box get the 36 Vdc via circuit breaker F3.

Fig. 13: Block diagram of the PDS



2.2 Function

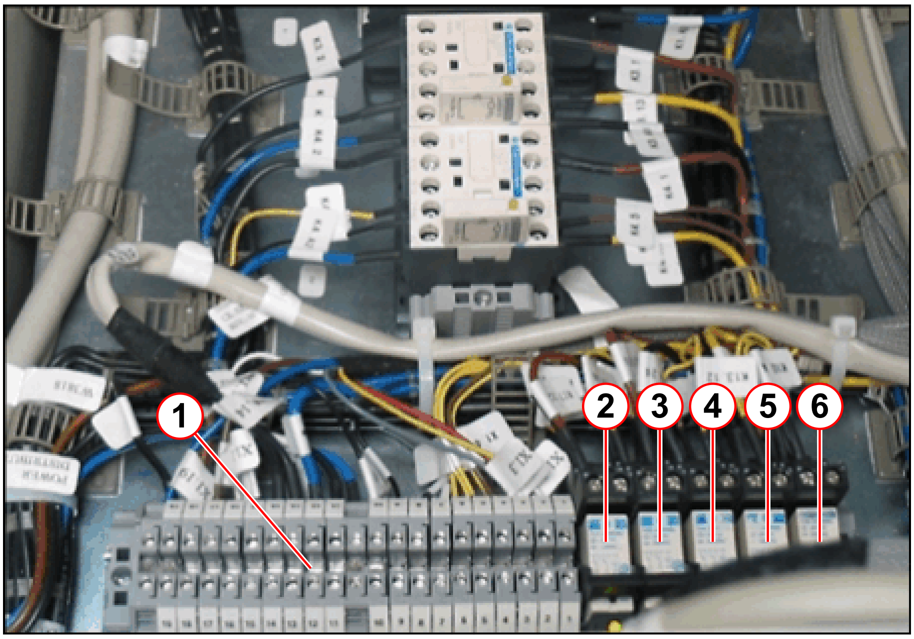
The power-up circuitry controls the system on, system off, and standby functionality of the power distribution system.

Fig. 14: 24/36 V power supply

The 24 Vdc and 36 Vdc power supply is located in the upper part of the ECA in front of the E-Shim device.

The dc relays of the power-up circuitry are next to the system contactors on a carrier plate above the main transformer T1.

Fig. 15: Carrier plate above main transformer T1



(1) X1 terminal block
(2) K17 (remote alarm)
(3) K15 (RFPA, SysCom, MARS)
(4) K13 (MRC in standby)
(5) K11 (system off)
(6) K10 (system on)

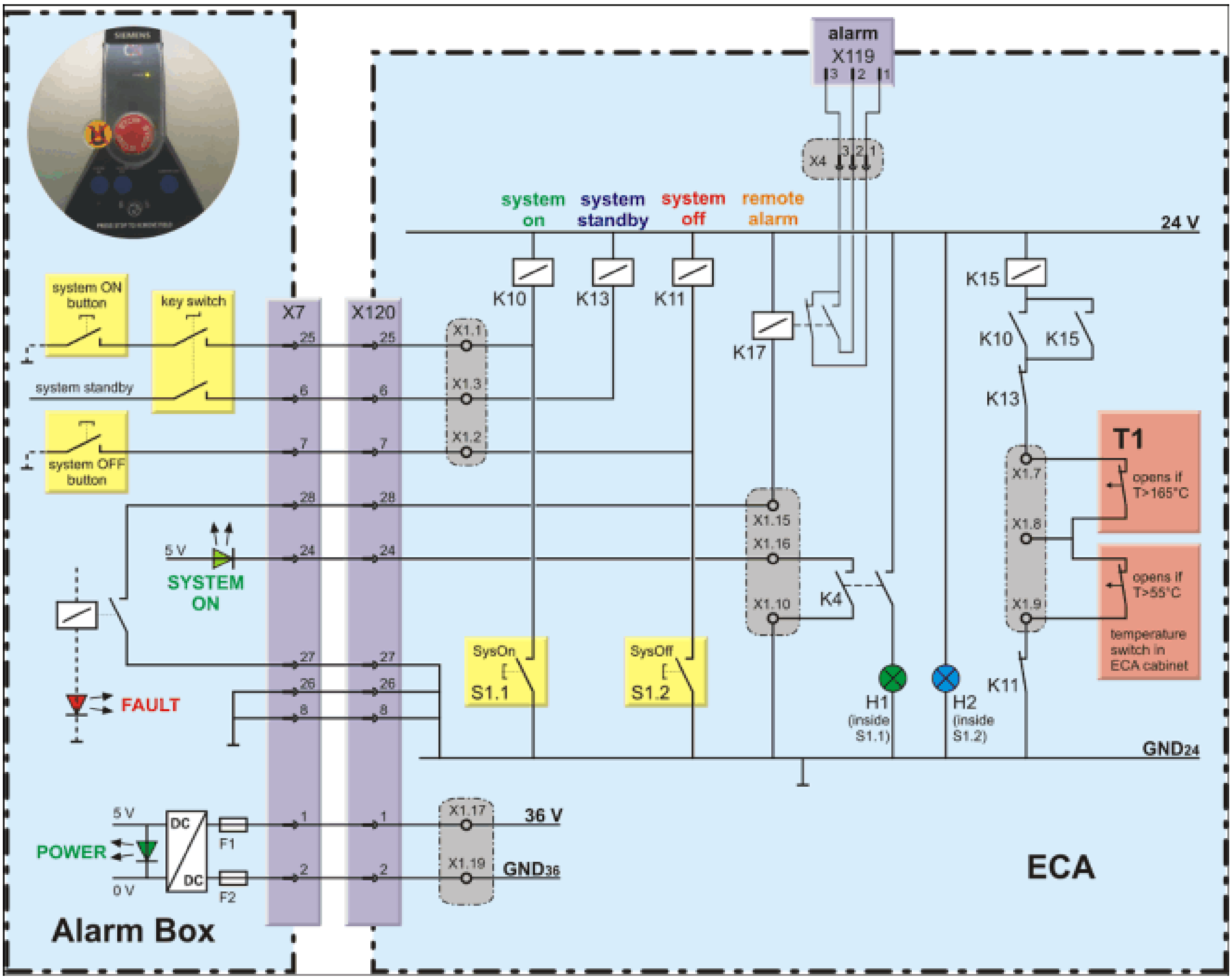
The following table shows the function of the dc relays:

Tab. 2 dc relays

dc relay	function
K10	energizes contactor K3 (MRC) and relay K15
K15	energizes contactor K1 (RFPA), K2 (SysComp), and K4 (MARS)
K11	de-energizes contactor K3 (MRC) and relay K15
K13	keeps contactor K3 energized (MRC) and de-energizes relay K15
K17	switches remote alarm connectors at X119 for customer alarm (FAULT-LED)

For details regarding the FAULT-LED in the Alarm Box and the alarm connector X119 on the ECA roof panel, refer to the Magnet System part, chapter Alarm Box.

Fig. 16: Power-up circuitry



2.2.1 System On

The MAGNETOM ESSENZA system can be switched on in two ways:

- SYSTEM ON button on the Alarm Box (key switch has to be in the unlock position)
- Button S1 ON at the right of the lower rows of circuit breakers inside the ECA.

> If the key switch is in the lock position and removed, you still can switch on the MAGNETOM ESSENZA system by using the S1 ON button in the PDS (inside ECA).

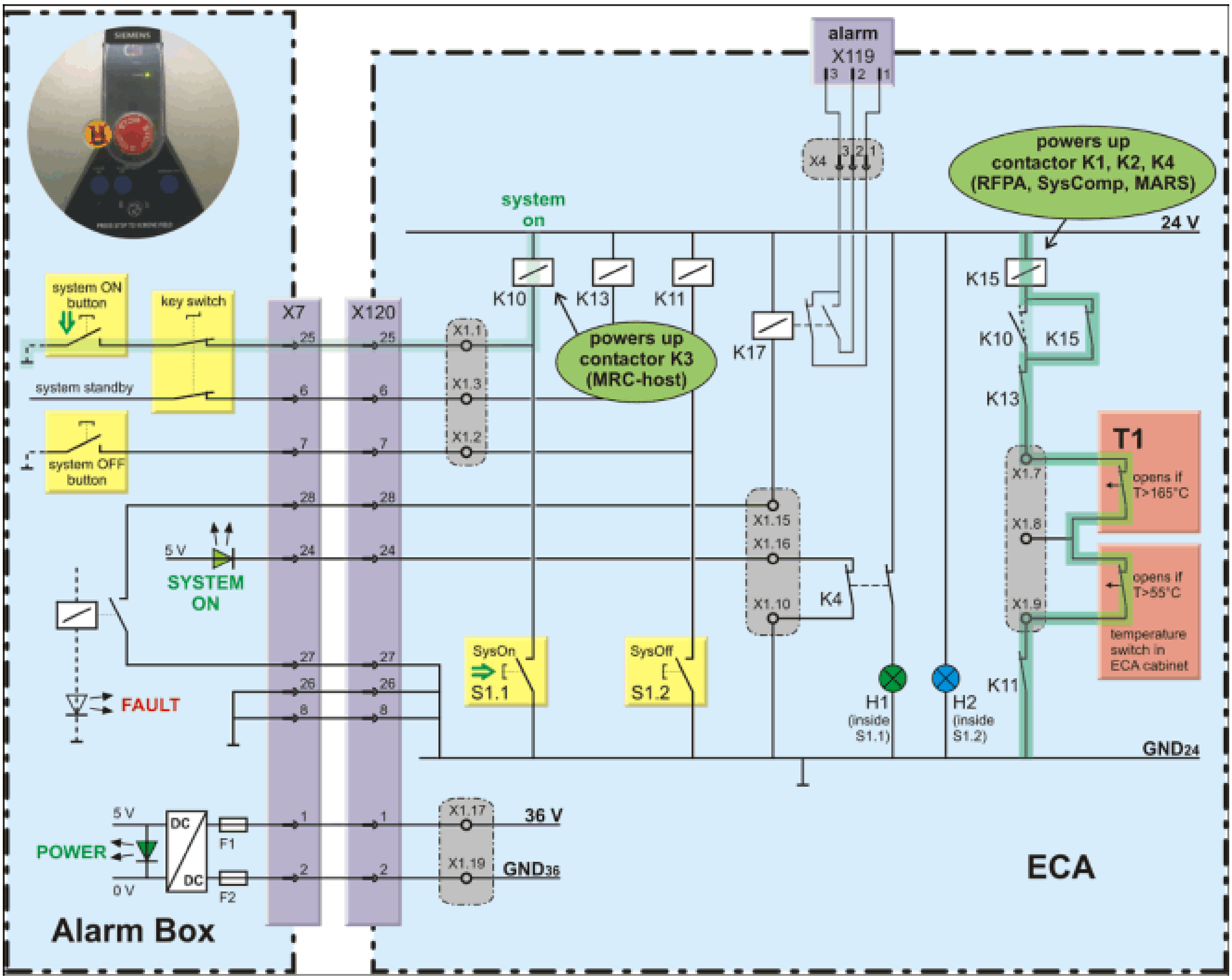
Fig. 17: System on

(1) SYSTEM ON at the Alarm Box
(2) S1 ON (with lamp H1 built in) at the PDS in the ECA

When SYSTEM ON is pressed

1. relay K10 is energized.
2. K10 energizes contactor K3 for MRC.
3. K10 energizes relay K15 if the temperature of the T1 main transformer is below 165˚C and the air temperature inside the ECA is below 55˚C.
4. relay K15 energizes contactors K1 (RFPA), K2 (SysComp), and K4 (MARS).
5. contactor K4 lights up the green SYSTEM ON LED at the Alarm Box and the green glow lamp in the S1 ON button in the ECA.

Fig. 18: Power-up circuitry (contacts shown in "system on" status)



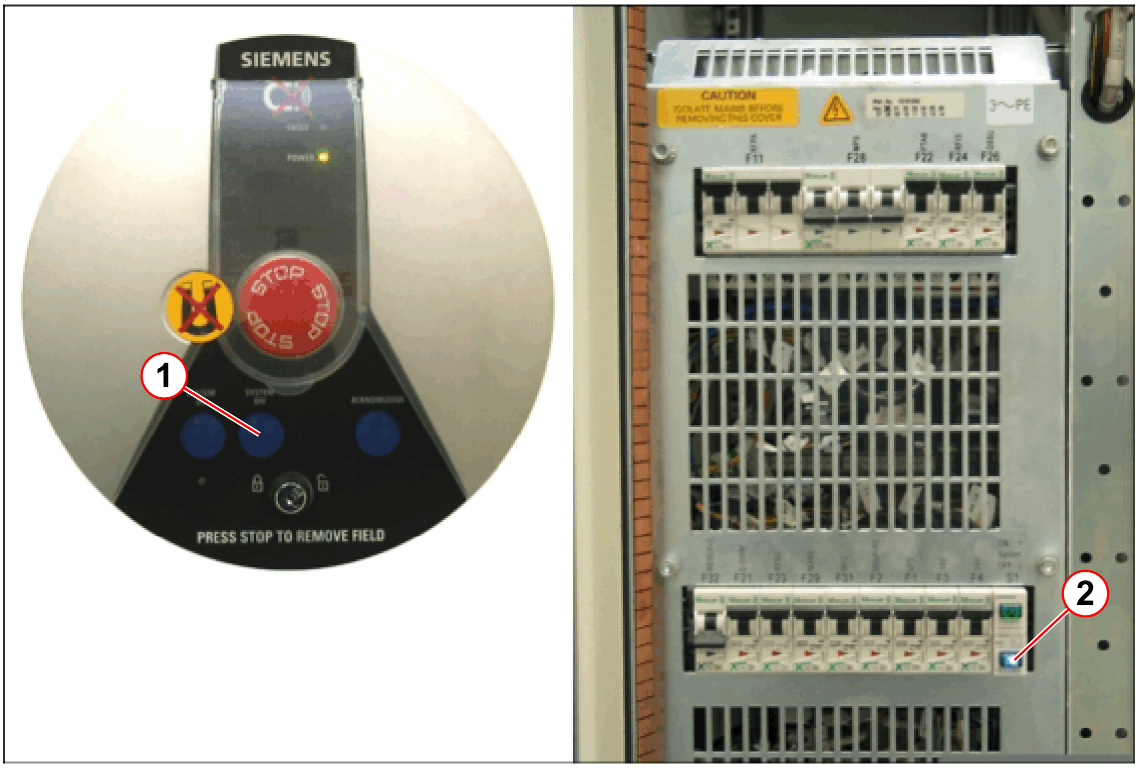
2.2.2 System Off

After shutdown of the NUMARIS application software and the operating system of the host computer, the power to the MAGNETOM ESSENZA system can be switched off in two ways:

- SYSTEM OFF button on the Alarm Box.
- Button S1 OFF at the right of the lower rows of circuit breakers inside the ECA.

If the MAGNETOM ESSENZA system is switched on, the key switch on the Alarm Box does not switch off the system when it is turned into the lock position.

Fig. 19: System off

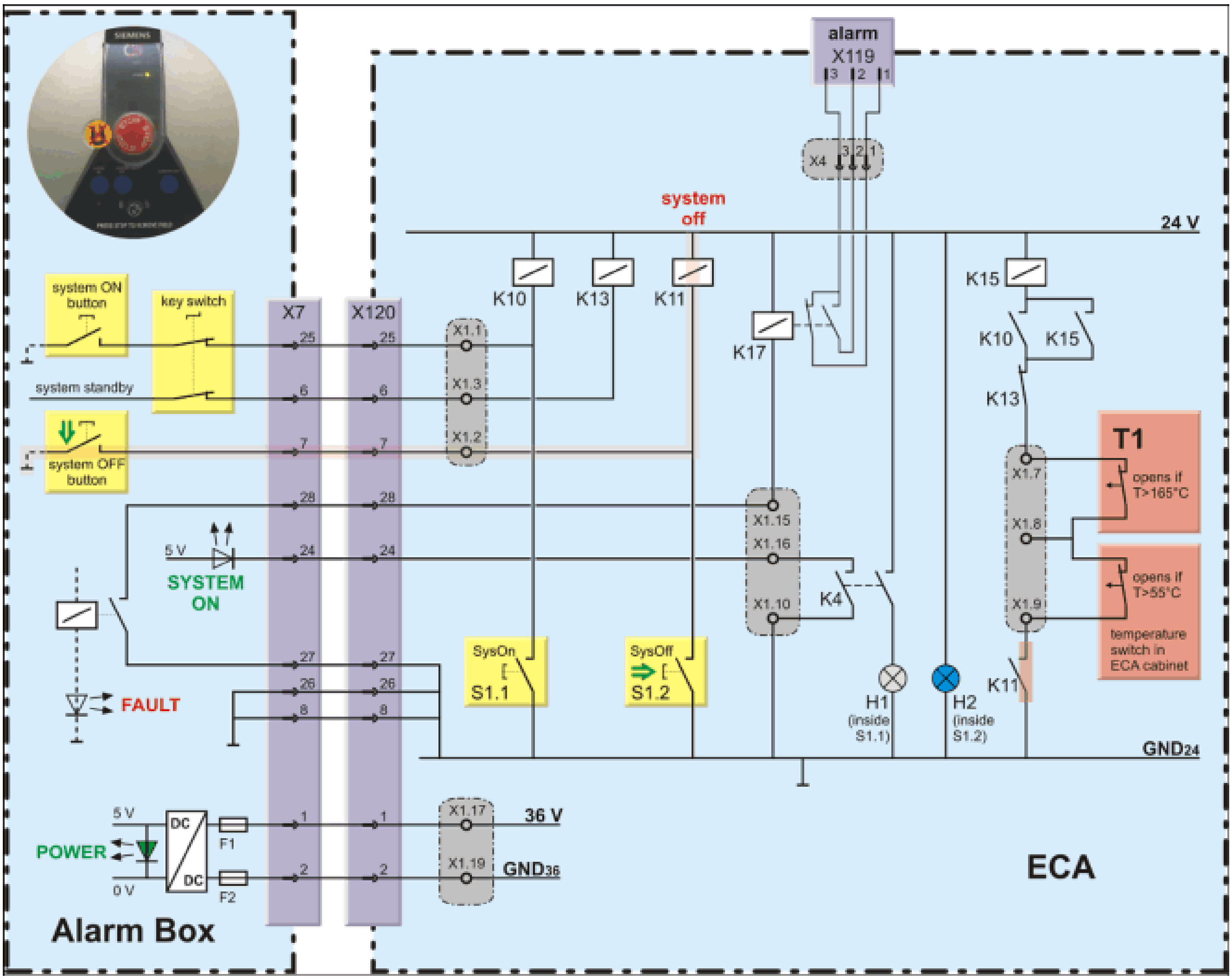


(1) SYSTEM OFF at the Alarm Box
(2) S1 OFF (with lamp H2 built in) at the PDS in the ECA

If SYSTEM OFF is pressed

1. relay K11 is energized.
2. K11 de-energizes contactor K3 for MRC.
3. K11 de-energizes relay K15.
4. if relay K15 drops, it de-energizes contactors K1 (RFPA), K2 (SysComp), and K4 (MARS).

Fig. 20: Power-up circuitry (contacts shown in "system off" status)



2.2.3 System Standby

The standby mode gives the user the ability to work with the MRC (e.g. for image evaluation, archiving, filming, etc.) while the rest of the system is switched off. This energy-saving mode can be used via software only. There is no button present on the Alarm Box as is the case on other systems.

The MAGNETOM ESSENZA system can be switched to standby mode in two ways:

- Main Menu > System > Control (System Manager) > Meas & Recon > Standby (➔ Fig. 21 Page 23)
- Main Menu > Options > Service > Local Service > Quality Assurance > General Quality Assurance > Standby (➔ Fig. 22 Page 23)

Fig. 21: System Manager

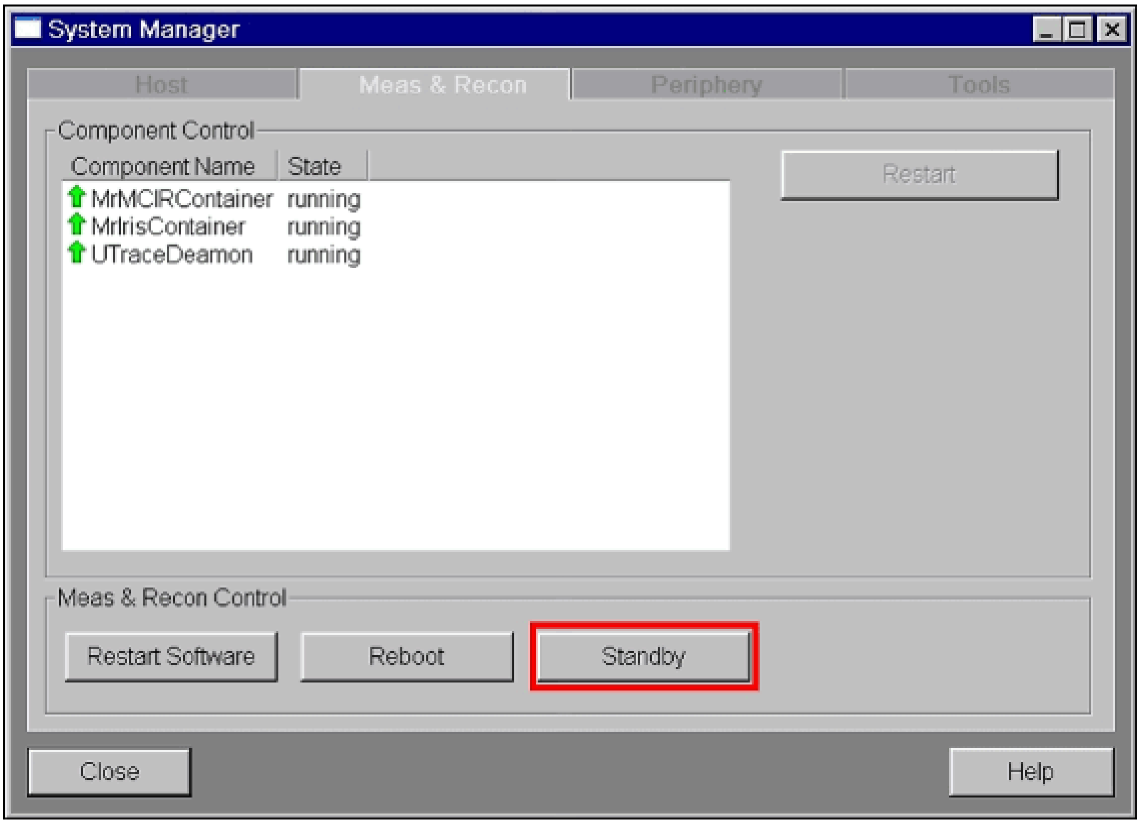


Fig. 22: General QA

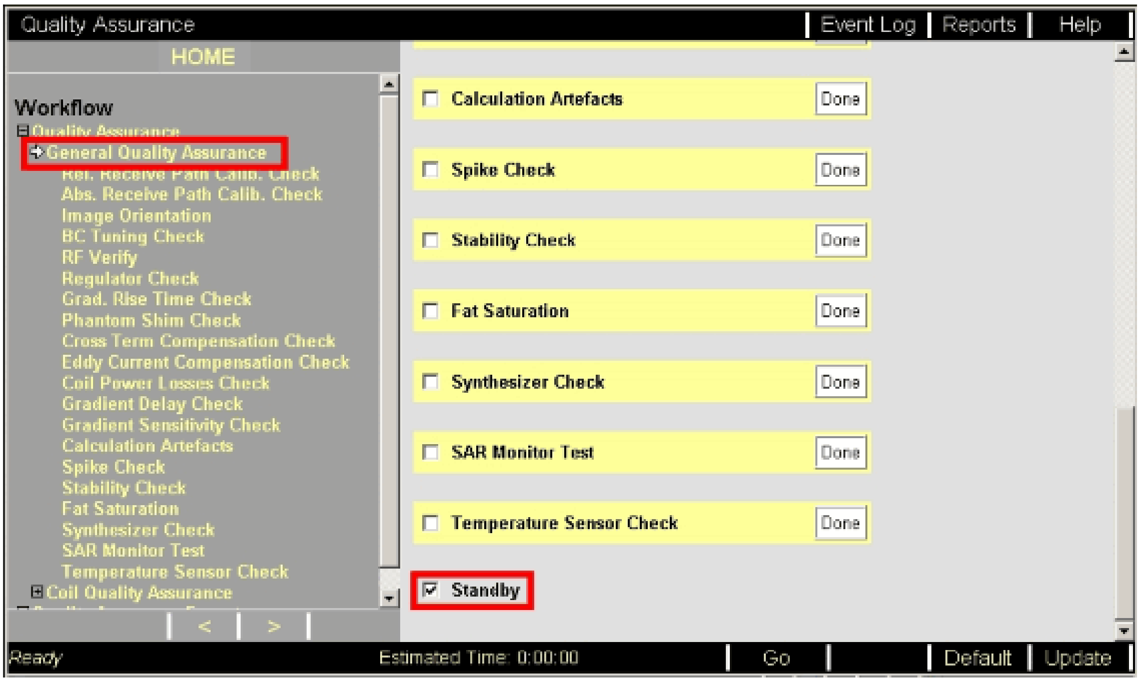
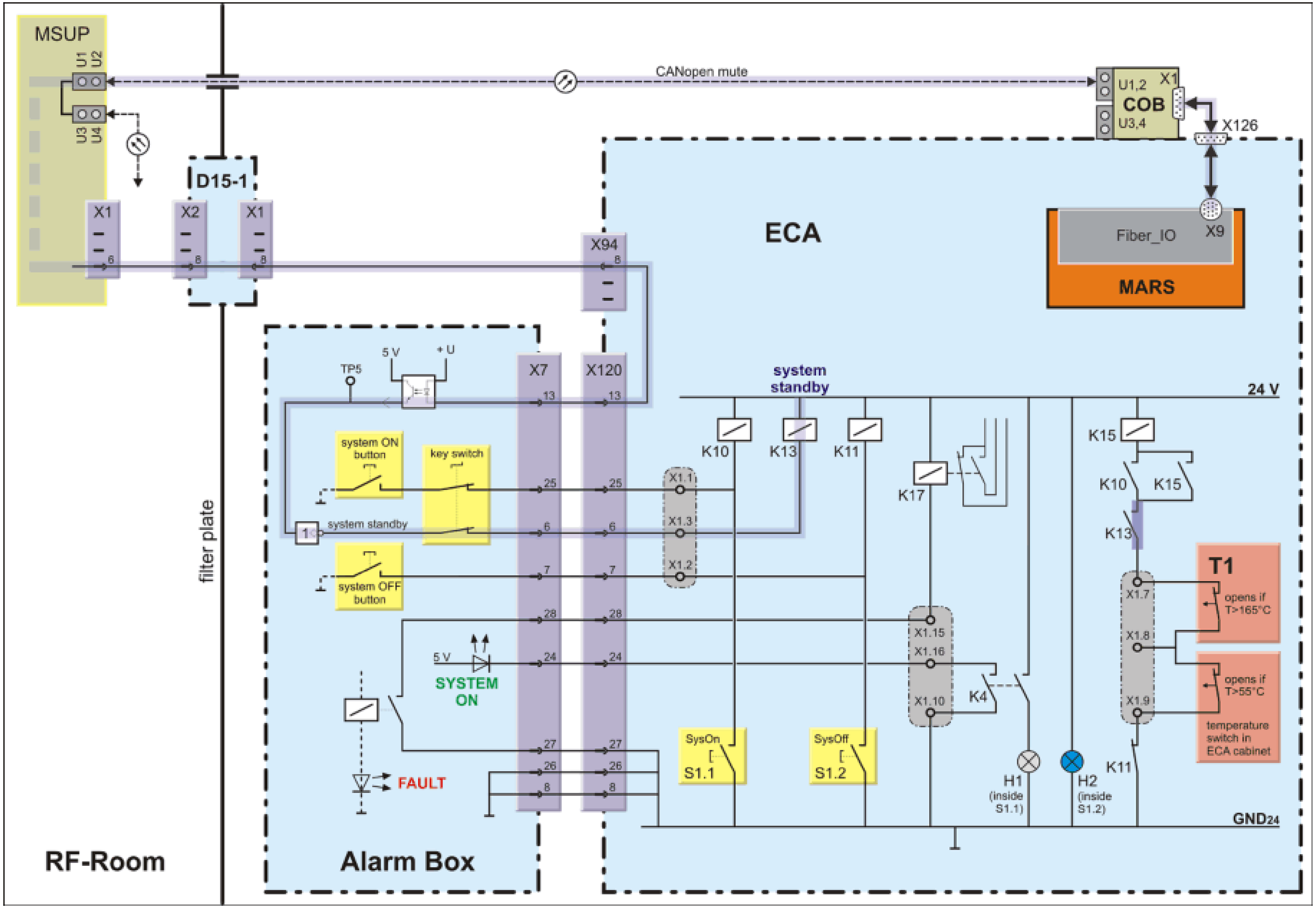


Fig. 23: Power-up circuitry (contacts shown in "system standby" status)



The key switch has to be in the unlock position to switch the MAGNETOM ESSENZA system into the standby mode.

If Standby is selected either in the System Manager or in the General Quality Assurance platform, the command is sent as a software instruction from the Host to the MARS using the ethernet link.

The MARS converts the command into a CAN-signal and sends it to the MSUP via the CAN-open Bridge (COB).

The MSUP converts the CAN-signal into a low-active switching signal sent via a dedicated signal line to the Alarm Box and from there to relay K13 in the PDS.

The system standby signal from the Alarm Box has the following effect:

1. relay K13 is energized.
2. K13 de-energizes relay K15.
3. if relay K15 drops, it de-energizes contactors K1 (RFPA), K2 (SysComp), and K4 (MARS).

If the system is in standby mode, it can be switched back to system-on mode by pressing SYSTEM ON on the Alarm Box or S1 ON on the PDS inside the ECA.

2.2.4 Temperature protection

The PDS consists of two temperature switches to protect against high temperatures during operation caused by failure or malfunction of the air-cooling or overload.

The two temperature switches are in the

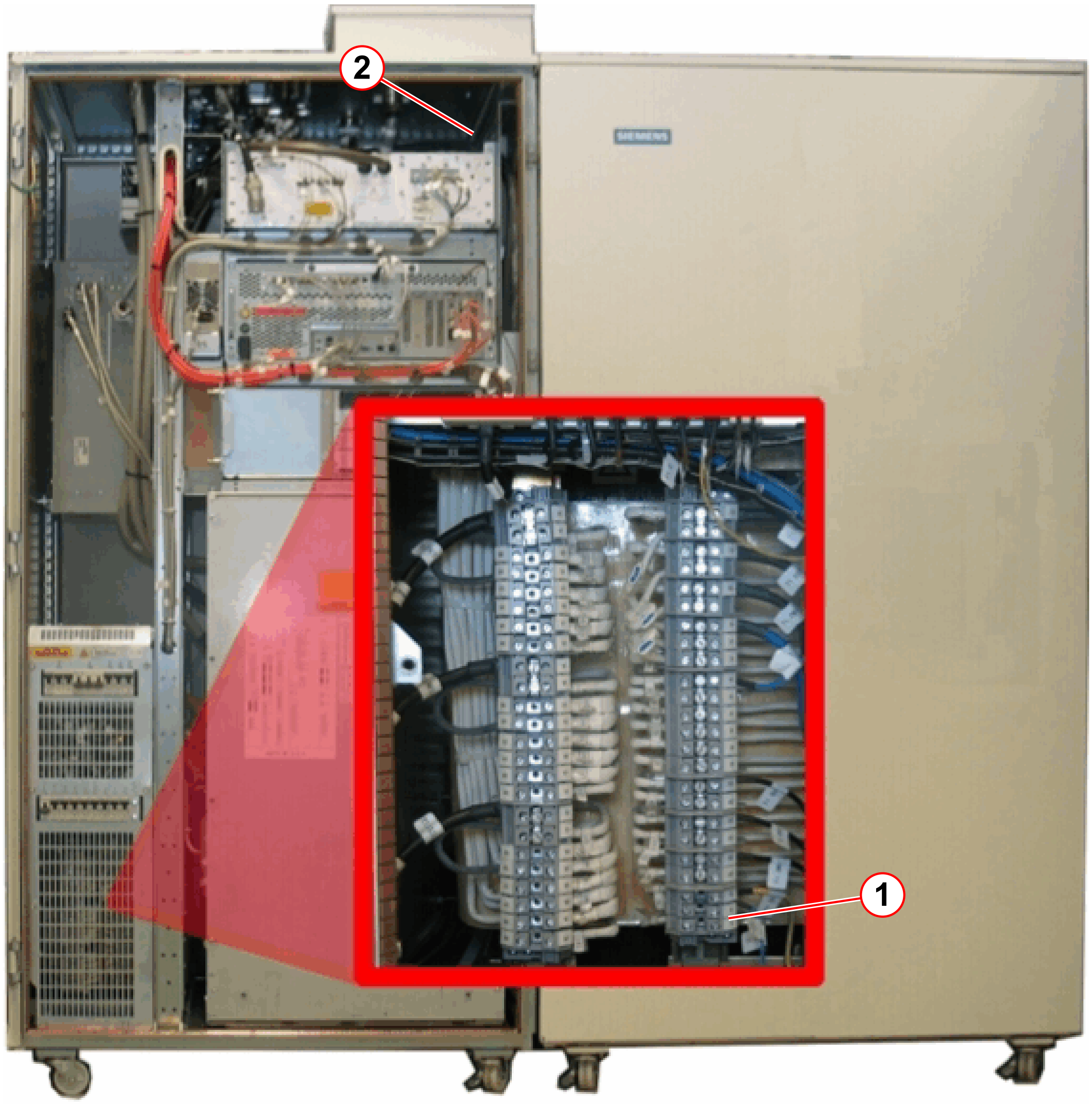
- main transformer T1 (temperature should be less than 165˚C)
- and in the upper part of the ECA, right above the RFPA (temperature should be less than 55˚C).

Both switches are connected in serial. If one opens, the result is the same as switching into standby mode:

1. At least one temperature switch is open;
2. this de-energizes relay K15;
3. if relay K15 drops, it de-energizes contactors K1 (RFPA), K2 (SysComp), and K4 (MARS).

If at least one of the temperature switches opens, the MRC remains on (as in standby mode).

Fig. 24: Position of the temperature switches



(1) Temperature switch in main transformer T1
(2) ECA air temperature switch

3 Changes to previous version

No previous version exists.

There are no Hazard IDs in this document.

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